Year 12 Maths - Probability

Similar questions - The binomial distribution

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A biased coin lands heads with probability 0.4. It is tossed 12 times and X is the number of heads.

(a) State two conditions needed for X to be modelled by a binomial distribution, and the distribution that results. **[2]**

(b) Find $\mathrm{P}(X = 5)$ and $\mathrm{P}(X \geqslant 3)$. **[3]**

(c) Find the smallest number of tosses n for which the probability of at least one head is greater than 0.99. **[3]**

(d) Without calculating each probability, explain why $\mathrm{P}(X = 5)$ is greater than $\mathrm{P}(X = 2)$. **[1]**

2 A manufacturer claims that 30% of its sweets are red. A pack contains 40 sweets and Y is the number of red sweets in a pack, modelled by $\mathrm{B}(40, 0.3)$.

(a) Find the expected number of red sweets in a pack and the probability that a pack contains exactly this number. **[3]**

(b) Find the largest value of n such that $\mathrm{P}(Y < n) < 0.05$. **[3]**

(c) Write down the sum of $\mathrm{P}(Y = r)$ for $r = 0, 1, \ldots, 40$, and explain how this is linked to the binomial expansion of $(0.7 + 0.3)^{40}$. **[2]**

Question 1

(a) A fixed number of independent tosses, each with the same probability of heads (two outcomes per toss). $X \sim \mathrm{B}(12, 0.4)$

(b) $\mathrm{P}(X = 5) = \binom{12}{5}(0.4)^5(0.6)^7 = 0.2270$. $\mathrm{P}(X \geqslant 3) = 1 - \mathrm{P}(X \leqslant 2) = 0.9166$

(c) $1 - 0.6^n > 0.99$, so $0.6^n < 0.01$ and $n > \dfrac{\ln 0.01}{\ln 0.6} = 9.01$. The smallest n is 10

(d) The mean is $12 \times 0.4 = 4.8$; 5 is much closer to the mean than 2 is, and binomial probabilities are largest near the mean

Question 2

(a) $\mathrm{E}(Y) = 40 \times 0.3 = 12$. $\mathrm{P}(Y = 12) = \binom{40}{12}(0.3)^{12}(0.7)^{28} = 0.1366$

(b) Using the cumulative tables/calculator: $\mathrm{P}(Y \leqslant 6) = 0.0238 < 0.05$ but $\mathrm{P}(Y \leqslant 7) = 0.0553 > 0.05$. So $\mathrm{P}(Y < 7) < 0.05$ and $n = 7$

(c) The probabilities sum to 1. $(0.7 + 0.3)^{40} = \sum \binom{40}{r}(0.3)^r(0.7)^{40-r}$, with each term equal to $\mathrm{P}(Y = r)$, and $(0.7 + 0.3)^{40} = 1^{40} = 1$